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Homework for Chap 16

For GaAs, the conduction band deformation potential $\Xi_c = 7.0\ eV$, and the valence band deformation potential $\Xi_c = -6.0\ eV$, and suppose the strain $\varepsilon = \Delta a/a = 10^{-3}$:

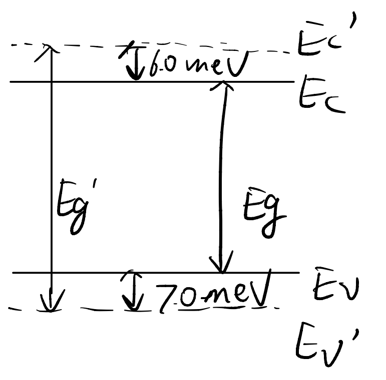
1. Calculate the change of the band gap energy ΔE_g;
2. Plot schematically how the conduction band and valence band shift compared with the unstrained GaA.

1. $\Delta E_c = 7.0\times10^{-3}\ eV$

$\Delta E_v = -6.0\times10^{-3}\ eV$

$$\Delta E_g = \Delta E_c - \Delta E_v = 1.3\times10^{-3}\ eV$$

2.



虚线代表 拉伸后
实线代表 拉伸前